[METRIC]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

HOSE, PREFORMED: SEMI-FLEXIBLE, REINFORCED

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers one type of semi-flexible preformed hose, for air induction on heavy duty engines.

2. SALIENT CHARACTERISTICS

2.1 Materials. Unless otherwise specified herein or in the applicable engineering drawing (AED) (see 6.2), the material used shall be in accordance with the manufacturer's material specifications and/or drawings. The use of recovered materials made in compliance with regulatory requirements is acceptable provided that all other requirements of this CID are met (see 3.1).

2.1.1 Rubber. Hose shall be made from rubber material in accordance with ASTM D2000.

2.1.2 Nylon. Nylon shall be in accordance with ASTM D4066.

2.1.3 Wire. Wire shall be in accordance with ASTM A818.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

FSC 4720

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2.2 Design and construction. Unless otherwise specified in the AED (see 6.2), the hose shall be designed and constructed in accordance with the manufacturers own drawings, specifications and standard practices.

2.3 Performance.

2.3.1 Bending. The hose shall be installed in a contour-forming fixture and subjected to a minimum radius bend of one and a half times the inside hose diameter for each of 4 separate directions spaced 90 degrees apart. The hose shall evidence no kinking, flattening, fracturing or deformation.

2.3.2 Collapsing resistance. A 15 centimeter (cm) (6 inch) hose sample shall evidence no damage in excess of 5% in diameter after being subjected to a load of 57 kilograms (kg) (125 pounds) uniformly applied for 5 minutes.

2.3.3 Accelerated-aging flexibility. The hose shall withstand the accelerated-aging test specified in ASTM D573 except the temperature shall be 121 ± 2 degrees Celsius (°C) (250 ± 3.6 degrees Fahrenheit (°F)) for a period of 24 ± .5 hours and then -54 ± 1 °C (-65 ± 2°F) for a period of 24 hours and flexed while at still latter temperature. After flexing, the hose shall withstand a pulsating proof pressure specified in 2.3.4 without any sign of damage.

2.3.4 Pulsating proof pressure. The samples shall withstand without any damage a pressure of 276 kilopascals (kPa) (40 pounds per square inch (psi)) applied internally at pulsating intervals of 5 seconds "on", 3 seconds "off" for 15 minutes. Sample shall be prepared in accordance with ASTM D380.

2.3.5 Burst strength. The hose shall show no evidence of bursting or leakage, at any pressure up to and including 517 kPa (75 psi), when subjected to a burst pressure specified in ASTM D380.

2.3.6 Fungus resistance. The hose shall be constructed of materials that will not support fungus growth. The sample shall not show any sign of damage when subjected to fungus growth as described in ASTM D3273 and then subjected to a burst pressure as described in paragraph 2.3.5.

2.3.7 Friction (adhesion). The hose shall withstand a force of at least 53 Newtons (N) (12 pounds), tube to ply. The sample shall be prepared in accordance with ASTM D413.

2.3.8 Resistance to vacuum. A 91.5 cm (36 inch) length of hose shall be assembled with end couplings. The outside diameter of the hose shall be measured and recorded. One end shall be closed in such a manner as to prevent leakage of air, and the other end shall be connected to a

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vacuum pump. The hose shall be subjected to a vacuum of 68 kPa (10 psi) or 50.8 cm of mercury (20 inches of mercury) while curved to a radius equal to 5 times the diameter of the hose. After a period of 5 minutes, and with vacuum maintained, the hose diameter shall again be measured and the hose inspected. When inspected the hose shall show no evidence of damage. The sample shall be prepared in accordance with ASTM D622.

2.4 Identification markings. Identification marking shall be placed on the hose. It shall be permanent and legible and shall include, as a minimum, the manufacturer's identification code (CAGE), manufacturer's part number and the part identification numbers (PIN) AA52546-1 (see 6.2).

2.5 Metal flanges. When metal flanges are specified (see 6.2), the hose shall be bonded to a metal "V" type flange on either or both ends and shall withstand a tensile load of 1335 N (300 pounds) without separating.

3. REGULATORY REQUIREMENTS

3.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

4.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.3 Market acceptability. The contractor shall provide products which have a proven market record based on the number of items sold, length of time the product has been on the market, and reliability and performance of the products as required under the contract or solicitation.

5. PACKAGING. Preservation, packing and marking shall be as specified in the contract or order (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Addresses for obtaining copies of referenced documents.

6.1.1 Non-Government documents. The following documents are available from the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103.

a. ASTM D2000 "Standard Classification System for Rubber Products in Automotive Applications".
b. ASTM D4066 "Standard Specification for Nylon Injection and Extrusion Materials (PA)".
c. ASTM A818 "Standard Specification for Coppered Carbon Steel Wire".
d. ASTM D573 "Standard Test Method for Rubber-Deterioration in an Air Oven".
e. ASTM D380 "Standard Test Methods for Rubber Hose".
f. ASTM D3273 "Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber".
g. ASTM D413 "Standard Test Methods for Rubber Property-Adhesion to Flexible substrate".
h. ASTM D622 "Standard Test Methods for Rubber Hose for Automotive Air and Vacuum Brake System".

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Applicable engineering drawing.
d. If design is other than as specified.
e. PIN number and quantity required.
f. Selection of applicable level and packaging requirements.
g. If metal flange is to be bonded on either or both ends.

6.3 Cross reference data. Preformed hose conforming to this CID are interchangeable/ substitutable with preformed hose conforming to MIL-H-52079B.

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PREPARING ACTIVITY:
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